

Functions

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DA-IICT

Chapter Summary

- 1 Introduction
- 2 Types of functions
- 3 Graph representation

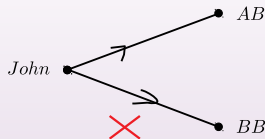
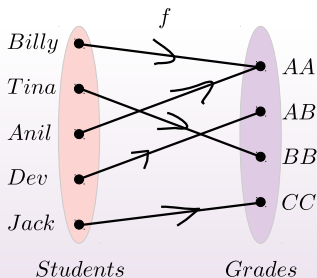
Functions

A and B are two non-empty sets.

Function: A function f from A to B is an assignment of each element of A to exactly one element of B .

$$f : A \rightarrow B$$

It is also called mapping or transformation.

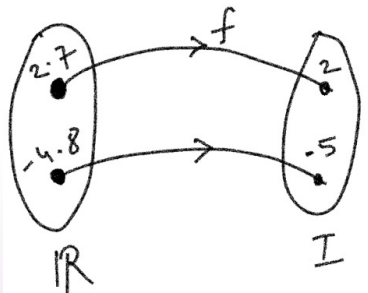


$$f(Dev) = AB$$

Example: The floor function

Example: The floor function

$$f : \mathbb{R} \rightarrow \mathbb{I}$$



$$b = f(a)$$

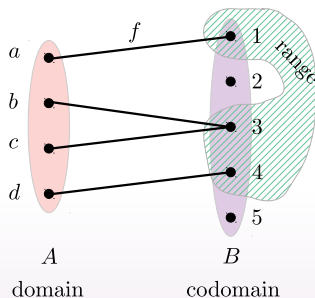
$$2 = f(2.7)$$

$$-5 = f(-4.8)$$

Terminologies

Domain, codomain, image, preimage, range of a function.

$$f : A \rightarrow B$$



A : Domain B : Codomain

$f(a) = 1$: f maps element $a \in A$ to element $1 \in B$.

1 is called the image of a and a is called the preimage of 1

Range of f is the set of all images of elements in A , i.e., $\{1, 3, 4\}$

Terminologies: example

Example: $f(x) = x^2$ is a function and $f : \mathbb{Z} \rightarrow \mathbb{Z}$

Domain: Set of all integers

Codomain: Set of all integers

Range: $\{0, 1, 4, 9, 16, \dots\}$

Addition and multiplication of real valued function

$$f_1 : A \rightarrow \mathbb{R}, \quad f_2 : A \rightarrow \mathbb{R}$$

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$(f_1 \cdot f_2)(x) = f_1(x) \cdot f_2(x)$$

Example: $f_1 : \mathbb{R} \rightarrow \mathbb{R}$ $f_2 : \mathbb{R} \rightarrow \mathbb{R}$

$$f_1(x) = x + 5 \quad f_2(x) = x^2 + x - 3$$

$$(f_1 + f_2)(x) = x + 5 + x^2 + x - 3$$

$$(f_1 \cdot f_2)(x) = (x + 5)(x^2 + x - 3)$$

Types of functions

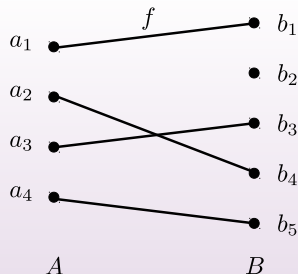
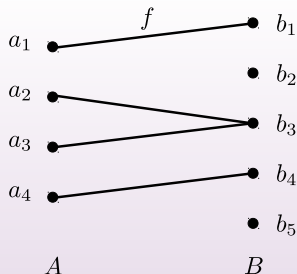
- Types of functions
 - One-to-one/injective
 - Onto/surjective
 - One-to-one and onto/bijective

One-to-one/injective

One-to-one/injective: A function f is called one-to-one/injective if each element in the codomain has a unique preimage.

$f : A \rightarrow B$ is one-to-one iff $f(x) = f(y)$ implies $x = y$ for every $x, y \in A$

f never assigns different elements in A to the same element in B



One-to-one/injective: example

Example: $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a function defined by $f(x) = x^2$. Is it one-to-one?

Solution: Here domain and codomain is the set of integers. $f(x) = x^2$
1 and -1 has the same image, that is, 1 has two preimages 1 and -1
4 has two preimages 2 and -2

Example: $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a function defined by $f(x) = x + 3$. Is it one-to-one?

Solution: Solution: We need to show that if $f(x) = f(y)$ then $x = y$

Assume that $f(x) = f(y)$, for $x, y \in \mathbb{Z}$

$f(x) = x + 3$ and $f(y) = y + 3$

$f(x) = f(y)$ implies that $x + 3 = y + 3 \rightarrow x = y$

One-to-one/injective: example

Consider the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by

$$f(x) = \begin{cases} 2x + 3, & \text{if } x \geq 0 \\ -x + 1, & \text{if } x < 0 \end{cases}$$

Is it one-to-one?

Solution: We need to show that if $f(x) = f(y)$ then $x = y$

Assume that $f(x) = f(y)$, for $x, y \in \mathbb{Z}$

There are three cases $x, y \geq 0$, $x, y < 0$, and $x \geq 0, y < 0$.

Case 1: $x, y \geq 0$ $f(x) = 2x + 3$ and $f(y) = 2y + 3$
 $f(x) = f(y)$ implies that $2x + 3 = 2y + 3 \implies x = y$

Case 2: $x, y < 0$ $f(x) = -x + 1$ and $f(y) = -y + 1$
 $f(x) = f(y)$ implies that $-x + 1 = -y + 1 \implies x = y$

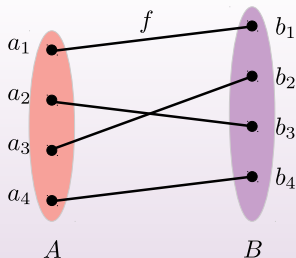
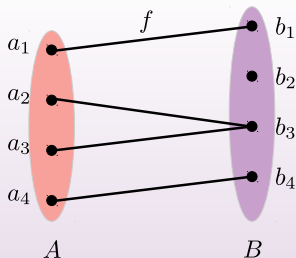
Case 3: $x \geq 0, y < 0$
 -6 and 2 has the same image.

Onto/surjective

Onto/surjective: A function f is called onto/surjective iff each element in the codomain is an image of some preimage.

$f : A \rightarrow B$ is onto iff every element $y \in B$, there is an element $x \in A$ such that $f(x) = y$

f never leaves any element in B unassigned, that is, range and codomain are the same.



One-to-one/injective: example

Example: $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a function defined by $f(x) = x^2$. Is it onto?

Solution: Here domain and codomain is the set of integers.

$$f(x) = x^2$$

2, 3 do not have preimages, i.e., no element in $\mathbb{Z}$ that maps $f(2)$ or $f(3)$

Example: $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a function defined by $f(x) = x + 3$. Is it onto?

Solution: For each element $y \in \mathbb{Z}$ there is an element x (i.e., $y - 3$) in $\mathbb{Z}$ such that $y = f(x)$

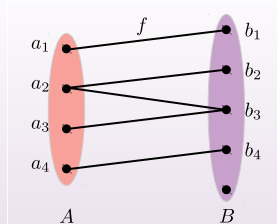
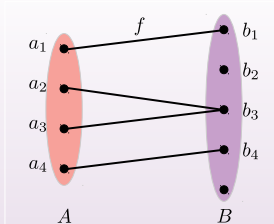
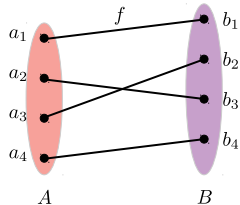
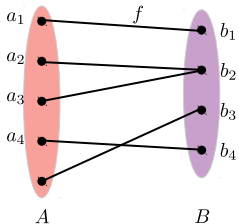
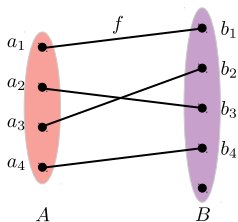
Bijjective (one-to-one and onto)

Bijjective: A function $f : A \rightarrow B$ is a bijective function if it is one-to-one and onto.

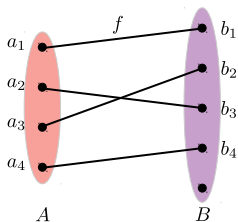
Example: $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a function defined by $f(x) = x + 3$. Is it bijective?

Solution: Need to prove f is both one-to-one and onto.

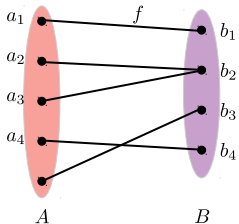
Functions visualization



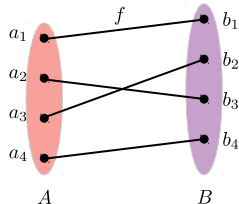
Functions visualization



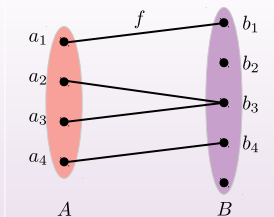
one-to-one but not onto



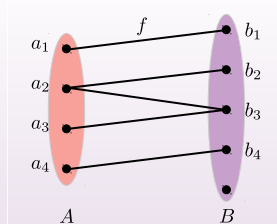
not one-to-one but onto



one-to-one and onto



neither one-to-one nor onto



not a function

Special function: identity function

A function where the image and preimage are the same.

$$f(x) = 1.x$$

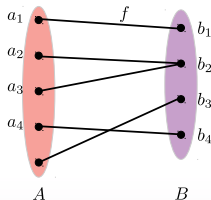
$$f(x) = x + 0$$

domain and codomain are the same set

Inverse functions

The inverse function is a function from B to A .

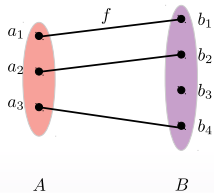
B is considered as a domain and A is considered as codomain.



$b_2 \in B$ have more than one images (a_2 and a_3).

Inverse will not defined a function.

To make it a function we must need f to be one-to-one



$b_3 \in B$ do not have any image.

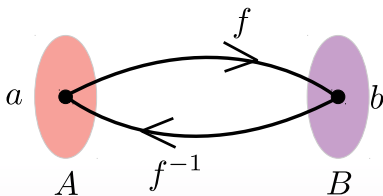
Inverse will not defined a function

To make it a function we must need f to be onto

Conclusion: In order to define inverse, f must be one-to-one and onto.

Inverse functions

The inverse of a bijective function f , written as f^{-1} , is the function that assigns to $b \in B$ a unique element $a \in A$ such that $f(a) = b$



Example: Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is bijective. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Inverse functions

Example: Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

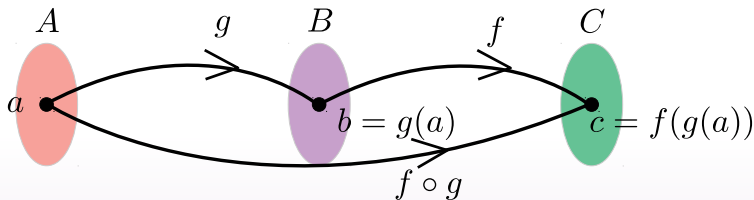
Solution:

- The function f has an inverse because it is bijective. why?
- Assume y is the image of x
- So that $y = x + 1$.
- Then $x = y - 1$.
- This means that $y - 1$ is the unique element of $\mathbb{Z}$ that is sent to y by f .
- Consequently, $f^{-1}(y) = y - 1$.

Composition of a function

Let $g : A \rightarrow B$ be a function and $f : B \rightarrow C$ be a function. Then the composition of f and g is denoted by $f \circ g$, is defined by:

$$(f \circ g)(x) = f(g(x))$$



Example: $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = 2x + 5$

$g : \mathbb{R} \rightarrow \mathbb{R}$ where $g(x) = x^2$

$$(f \circ g)(2) = f(g(2)) = f(4) = 13$$

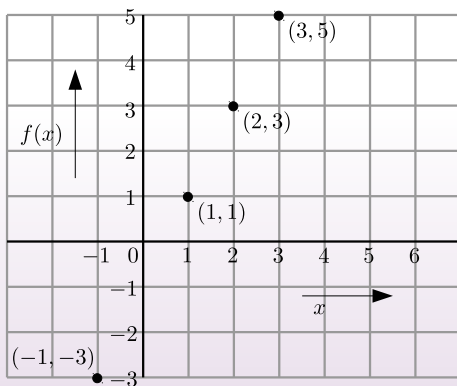
$$(g \circ f)(2) = g(f(2)) = g(9) = 81$$

Conclusion: $(f \circ g) \neq (g \circ f)$

Graph representaion of a function

$f : A \rightarrow B$ be a function, then the graph representation of f is the set of ordered pairs $\{(a, b) | a \in A, b \in B\}$ where $f(a) = b$

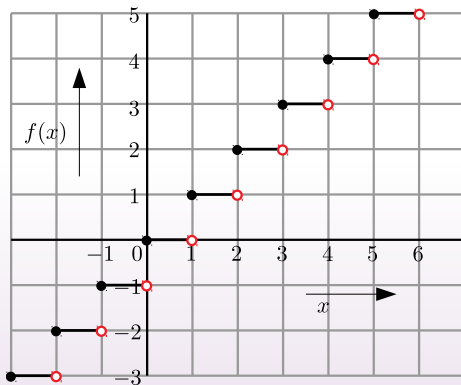
Example: $f : \mathbb{Z} \rightarrow \mathbb{Z}$ where $f(x) = 2x - 1$



Floor function

$f : \mathbb{R} \rightarrow \mathbb{Z}$ defined by

$f(x) = \lfloor x \rfloor$, where $\lfloor x \rfloor$ is the largest integer less than or equal to x

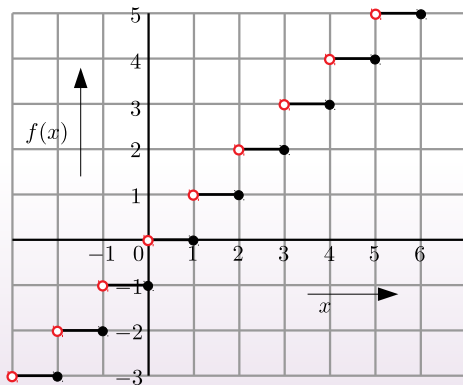


Here, hollow circular point is not included.

Celing function

$f : \mathbb{R} \rightarrow \mathbb{Z}$ defined by

$f(x) = \lceil x \rceil$, where $\lceil x \rceil$ is the smallest integer greater than or equal to x



Here, hollow circular point is not included.